Applied Plotting, Charting & Data Representation in Python

by University of Michigan

About this Course

This course will introduce the learner to information visualization basics, with a focus on reporting and charting using the matplotlib library. The course will start with a design and information literacy perspective, touching on what makes a good and bad visualization, and what statistical measures translate into in terms of visualizations. The second week will focus on the technology used to make visualizations in python, matplotlib, and introduce users to best practices when creating basic charts and how to realize design decisions in the framework. The third week will be a tutorial of functionality available in matplotlib, and demonstrate a variety of basic statistical charts helping learners to identify when a particular method is good for a particular problem. The course will end with a discussion of other forms of structuring and visualizing data. This course should be taken after Introduction to Data Science in Python and before the remainder of the Applied Data Science with Python courses: Applied Machine Learning in Python, Applied Text Mining in Python, and Applied Social Network Analysis in Python.

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Taught by:

Christopher Brooks
Associate Professor, School of Information

Basic Info	Course 2 of 5 in the Applied Data Science with Python Specialization
Level	Intermediate
Language	English Subtitles: Arabic, French, Bengali, Ukrainian, Chinese (Simplified), Greek, Italian, Portuguese (Brazil), Vietnamese, Dutch, Korean, German, Pashto, Urdu, Russian, Thai, Indonesian, Swedish, Turkish, Azerbaijani, Spanish, Dari, Hindi, Japanese, Kazakh, Hungarian, Polish
How To Pass	Pass all graded assignments to complete the course.
User Ratings	★ 4.5

Syllabus

Module 1

Module 1: Principles of Information Visualization

In this module, you will get an introduction to principles of information visualization. We will be introduced to tools for thinking about design and graphical heuristics for thinking about creating effective visualizations. All of the course information on grading, prerequisites, and expectations are on the course syllabus, which is included in this module.

8 videos, 6 readings

1. Video: Introduction
2. Video: Updates
3. Reading: Syllabus
4. Reading: Help us learn more about you!
5. Video: About the Professor: Christopher Brooks
6. Video: Tools for Thinking about Design (Alberto Cairo)
7. Reading: Notice for Coursera Learners: Assignment Submission
8. App Item: Hands-on Visualization Wheel
9. Video: Graphical heuristics: Data-ink ratio (Edward Tufte)
10. Reading: Dark Horse Analytics (Optional)
11. Video: Graphical heuristics: Chart junk (Edward Tufte)
12. Reading: Useful Junk?: The Effects of Visual Embellishment on Comprehension and Memorability of Charts
13. Video: Graphical heuristics: Lie Factor and Spark Lines (Edward Tufte)
14. Video: The Truthful Art (Alberto Cairo)
15. Discussion Prompt: Must a visual be enlightening?
16. Reading: Graphics Lies, Misleading Visuals

Graded: Graphics Lies, Misleading Visuals

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Module 2

Module 2: Basic Charting

In this module, you will delve into basic charting. For this week's assignment, you will work with real world CSV weather data. You will manipulate the data to display the minimum and maximum temperature for a range of dates and demonstrate that you know how to create a line graph using matplotlib. Additionally, you will demonstrate the procedure of composite charts, by overlaying a scatter plot of record breaking data for a given year.

7 videos, 2 readings

1. Ungraded Lab: Module 2 Jupyter Notebooks
2. Video: Introduction
3. Video: Matplotlib Architecture
4. Reading: Matplotlib
5. Reading: Ten Simple Rules for Better Figures
6. Video: Basic Plotting with Matplotlib
7. Video: Scatterplots
8. Video: Line Plots
9. Video: Bar Charts
10. Video: Dejunkifying a Plot
11. Ungraded Lab: Plotting Weather Patterns

Graded: Plotting Weather Patterns

Module 3

Module 3: Charting Fundamentals

In this module you will explore charting fundamentals. For this week's assignment you will work to implement a new visualization technique based on academic research. This assignment is flexible and you can address it using a variety of difficulties - from an easy static image to an interactive chart where users can set ranges of values to be used.

6 videos, 3 readings

1. Ungraded Lab: Module 3 Jupyter Notebooks
2. Video: Subplots
3. Video: Histograms
4. Reading: Selecting the Number of Bins in a Histogram: A Decision Theoretic Approach (Optional)
5. Video: Box Plots
6. Video: Heatmaps
7. Video: Animation
8. Video: Widget Demonstration
9. Ungraded Lab: Practice Assignment: Understanding Distributions Through Sampling
10. Peer Review: Practice Assignment: Understanding Distributions Through Sampling
11. Ungraded Lab: Building a Custom Visualization
12. Reading: Assignment Reading
13. Reading: Understanding Error Bars

Graded: Building a Custom Visualization

Module 4

Module 4: Applied Visualizations

In this module, then everything starts to come together. Your final assignment is entitled "Becoming a Data Scientist." This assignment requires that you identify at least two publicly accessible datasets from the same region that are consistent across a meaningful dimension. You will state a research question that can be answered using these data sets and then create a visual using matplotlib that addresses your stated research question. You will then be asked to justify how your visual addresses your research question.

4 videos, 3 readings

1. Ungraded Lab: Module 4 Jupyter Notebooks
2. Video: Plotting with Pandas
3. Video: Seaborn
4. Reading: Spurious Correlations
5. Video: Mapping and Geographic Investigation
6. Video: Becoming an Independent Data Scientist
7. Ungraded Lab: Project Description
8. Reading: Post-course Survey
9. Reading: 5 reasons to keep going

Graded: Becoming an Independent Data Scientist

How It Works

General

What do start dates and end dates mean?
Once you enroll, you'll have access to all videos, readings, quizzes, and programming assignments (if applicable). If you choose to explore the content without purchasing, you may not be able to access certain assignments. If you don't finish all graded assignments before the end of the session, you can reset your deadlines. Your progress will be saved and you'll be able to pick up where you left off.

What are due dates? Is there a penalty for submitting my work after a due date?
Within a course, there are suggested due dates to help you manage your schedule and keep work from piling up. Quizzes and programming assignments can be submitted late without consequence. However, it is possible that you won't receive a grade if you submit your peer-graded assignment too late because classmates usually review assignment within three days of the assignment deadline.

Can I re-attempt an assignment?
Yes. If you want to improve your grade, you can always try again. If you're re-attempting a peer-graded assignment, re-submit your work as soon as you can to make sure there's enough time for your classmates to review your work. In some cases you may need to wait before re-submitting a programming assignment or quiz. We encourage you to review learning material during this delay.

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Peer-graded assignments

Peer-graded assignments require you and your classmates to grade each other's work.

How do peer graded assignments work?
After you submit your assignment, you will review some of your peers' assignments. The number of assignments you must review is set by the instructor of the course.

I reviewed my peers' assignments! What happens next?
While you're reviewing your peers' assignments, they'll review yours. If you submit your assignment on time, you'll get your grade within a week, as long as at least one peer reviews your assignment. If you submit late, you'll need all of the peer reviews the instructor requires. Learn more about Peer Graded Assignments.

How are grades calculated?
You and your classmates will be asked to provide a score for each part of the assignment. Final grades are calculated by combining the median scores you received for each section.

What kind of feedback should I give?
Use the instructor's criteria in the rubric to grade honestly and fairly. If your peers' answers are excellent, score them highly and tell them what they did well. If their answers aren't as good, give the score they deserve, and be sure to provide respectful, useful feedback so they can do better next time they attempt the assignment.

Is there a penalty for submitting my work late?
No, but it's important to submit your work as close to the due date as you can. Classmates grade most of the assignments within three days of the due date. If you submit yours too late, there may not be anyone to review your work.

If I fail an assignment, can I try again?
Yes! You can always try again, but you'll need to resubmit your work as soon as possible to make sure your classmates have enough time to grade your work.

Can I edit my assignment?
Yes, but you'll need to re-submit your work and any grade you've already received will be deleted.

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